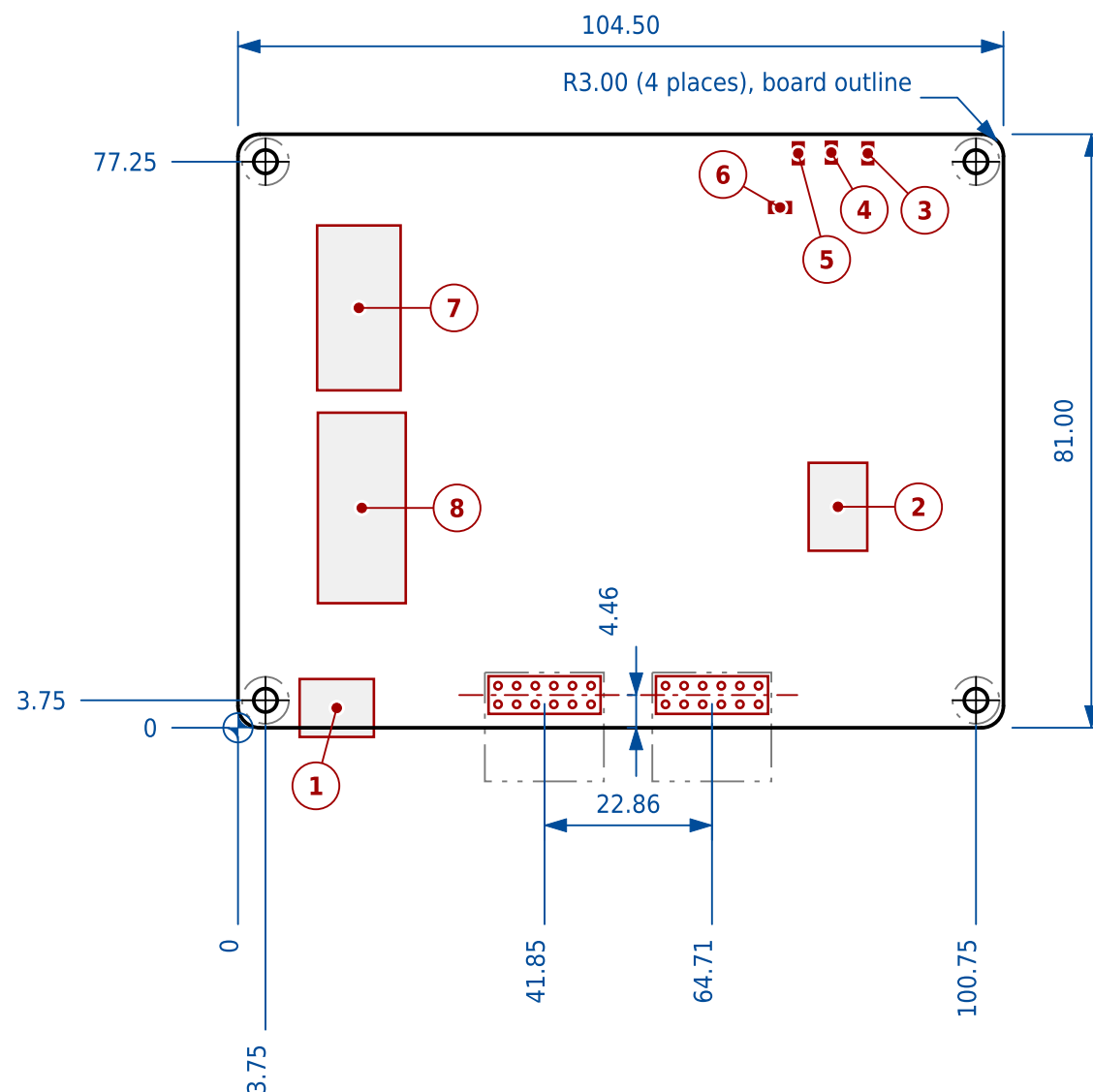




## NOTES

1. Viewed from the component side.
2. Ships in TT02 Dev Kit and TT03 Dev Kit.
3. Hole IDs are the KiCad reference designators.
4. Named "TinyTapeout 1,2,3 Demo Board" in KiCad, but no PCB served TT01, which was a bare-die trial run.
5. The revisions covered are mechanically identical; only the electrical design differs.
6. Pmod host pitch, 22.86 mm (0.9 in), is the Digilent Pmod Interface Specification 1.2.0 figure and is the same on every revision; the mounting plate, TT-MP-01, registers against it.
7. Assembled envelope, connector overhang included: 104.50 x 82.25 mm.

## SOURCES

1. KiCad board file:  
<https://github.com/TinyTapeout/tt123-demo-pcb>  
mpw-mb1.kicad\_pcb @ 303509a - title block: TinyTapeout 1,2,3 Demo Board rev 2.2.5, dated 2023-11-13
2. KiCad board file:  
<https://github.com/TinyTapeout/tt123-demo-pcb>  
mpw-mb1.kicad\_pcb @ 3d721a6 - title block: TinyTapeout 1,2,3 Demo Board rev 2.2.6, dated 2024-02-14
3. Shuttle mapping: <https://mith.ro/tt-demoboards/> - used by: TT02, TT03

## HOLE SCHEDULE

| ID  | X mm   | Y mm  | DIA mm | KEEPOUT mm |
|-----|--------|-------|--------|------------|
| MT1 | 3.75   | 3.75  | 3.20   | 6.40       |
| MT2 | 100.75 | 3.75  | 3.20   | 6.40       |
| MT3 | 3.75   | 77.25 | 3.20   | 6.40       |
| MT4 | 100.75 | 77.25 | 3.20   | 6.40       |

## PMOD HOST HEADERS

| PORT   | REF | EDGE   | CX mm  | CY mm | PIN1 X mm | PIN1 Y mm |
|--------|-----|--------|--------|-------|-----------|-----------|
| PMOD A | J3  | bottom | 41.850 | 4.465 | 48.200    | 5.735     |
| PMOD B | J9  | bottom | 64.710 | 4.465 | 71.060    | 5.735     |

## FEATURE SCHEDULE

| #  | FEATURE                                        | X EXTENT mm    | Y EXTENT mm    |
|----|------------------------------------------------|----------------|----------------|
| 1  | USB-C power / control                          | 8.41 to 18.55  | -1.25 to 6.65  |
| 2  | 7-segment display                              | 77.91 to 85.91 | 24.16 to 36.16 |
| 3  | LED D1                                         | 85.25 to 86.72 | 76.92 to 79.88 |
| 4  | LED D2                                         | 80.27 to 81.73 | 77.02 to 79.98 |
| 5  | LED D3                                         | 75.75 to 77.22 | 76.92 to 79.88 |
| 6  | LED D4                                         | 72.52 to 75.48 | 70.27 to 71.73 |
| 7  | 8-way input DIP switch                         | 10.80 to 22.20 | 46.05 to 68.55 |
| 8  | 9-way DIP switch SW2                           | 10.90 to 22.90 | 17.00 to 43.00 |
| 9  | Side Pmod, first position - not on this board  | -              | -              |
| 10 | Side Pmod, second position - not on this board | -              | -              |
| 11 | Side Pmod, third position - not on this board  | -              | -              |

## LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body, hole keep-out
- Dimension, extension and leader

## DB MPW V2.2.5 / DB MPW V2.2.6

mpw-mb1 rev 2.2.5 and 2.2.6

|                  |                   |                                                |       |
|------------------|-------------------|------------------------------------------------|-------|
| DRAWN            | VERSION           | UNITS                                          | SCALE |
| generated        | v0.0-122-gf3d0d0b | mm                                             | 1:1   |
| SIZE             | SHEET             | DRAWING NO                                     | REV   |
| A3               | 1 OF 1            | TT-DB-01                                       | A     |
| MATERIAL         |                   | GENERAL TOLERANCE                              |       |
| PCB, 1.6 nominal |                   | edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08 |       |